Angelos Ioannis Katsampekis

Senior Software Engineer | Unity, Physics, XR, AI & Simulation

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PROFILE

Senior Software Engineer with 6+ years building simulation, XR and AI systems in Unity, C#, C++, Python, and Rust. At **Micro1**, building AI training-data pipelines: a Godot game-engine pipeline for AI coding-agent evaluation, and single-player and multiplayer game capture pipelines (game mods, DX11/DX12/Vulkan hooks) for **foundation world models and robotics**. Lead Unity development of a modular real-time 3D exhibition platform. Delivered projects for **Samsung, Coca-Cola and EU Horizon** programs, ranging from UAV/missile simulation and reinforcement-learning navigation to mobile AI deployment and XR tracking systems.

SKILLS

Programming
Python, C#, C++, Rust, JavaScript, Lua, WitcherScript, REST APIs, JSON

Game Engines & Simulation
Unity (URP, HDRP, DOTS, Jobs System, Burst Compiler), Godot, game modding, physics simulation, aerodynamics, multithreading, performance profiling and optimization

AI Pipelines & Evaluation
AI training data pipelines, world model training data, synthetic data generation, multiplayer data capture, LLM evaluation, image evaluation, AI coding-agent evaluation, code review, data annotation, evaluation task and acceptance-criteria design, automated verifiers and graders, quality assurance (QA), benchmarking

Machine Learning
Deep reinforcement learning (RL), Unity ML-Agents, ONNX Runtime, TensorFlow Lite, NNAPI, Unity Sentis, generative AI integration

XR / VR / AR
Meta Quest, Valve Index, OpenXR, Meta XR SDKs

Graphics
DX11, DX12, Vulkan, OpenGL, Graphics API Hooking, Frame Capture, Render-State Inspection, Camera/View/Projection Matrix Extraction, Intrinsics, Depth

Tools & Networking
Docker, Linux, WSL, Git, CI/CD, Netcode for GameObjects, 5G video streaming, Cesium 3D Tiles, geospatial data

Ways of Working
Technical documentation, process design, stakeholder communication, client reporting, crossfunctional collaboration, remote B2B delivery

PROFESSIONAL EXPERIENCE

Senior Software Engineer
Micro1
03/2026 – Present | Remote
- Game-engine pipeline (Godot): built a pipeline for AI coding-agent game tasks and evaluation; design tasks with acceptance criteria, review agent-generated code and debug automated graders (Python, Rust, Docker, Linux).
- Game data capture pipeline: built game mods and DX11/DX12/Vulkan/OpenGL hooks that extract frames, camera matrices, intrinsics, depth and semantic actions from 3D games such as Cyberpunk 2077, Crimson Desert and RDR2 as synthetic training data for embodied AI and robotics.
- Multiplayer capture: extended the pipeline to multiplayer with dedicated-server telemetry (Minecraft, Arma 3, Mordhau), synchronized client/server events and sessions of up to 64 players.

AI Trainer
Micro1
10/2025 – 03/2026
Evaluated and improved AI model outputs on coding, game-development and image evaluation tasks, writing structured feedback and quality ratings used as training data. Promoted to Senior Software Engineer.

Head of Unity Development
Institution Srl
02/2026 – Present
- Lead Unity development of a modular 3D platform for virtual art pavilions, spatial exhibitions, and online cultural participation.

Software Developer
Pragma – IoT Solutions
01/2025 – 08/2025 | Thessaloniki, Greece
- Built Python and C++ tools that convert and package ONNX models for Android NNAPI, streamlining on-device AI deployment for Samsung.

Unity Developer – Research Assistant
CERTH – The Centre for Research & Technology, Hellas
11/2022 – 08/2025 | Thessaloniki, Greece
- Developed XR, simulation, geospatial, AI, and 5G streaming systems across EU Horizon and industry research projects, working directly with partner teams including Samsung Electronics and Coca-Cola HBC.

Unity Developer - XR & AI Simulation Engineer
Self-Employed (Freelance)
2020 – Present
Deliver Unity simulation, XR and AI projects for clients, from concept and architecture to deployment.

EDUCATION

Physics (BSc)
Aristotle University of Thessaloniki
2020

Advanced Computer and Communication Systems (MSc)
Aristotle University of Thessaloniki
2025

Data Science with Python (Certificate)
National and Kapodistrian University of Athens
2021

LANGUAGES

- Greek
- English

SELECTED PROJECTS

Drone & Missile Simulator (Unity)

- Designed and built a complete **physics-based flight simulator** end-to-end, from core architecture and aerodynamics modeling to UX and control systems.
- Developed and optimized a custom **multithreaded aerodynamic solver** with per-vertex force computation, parallelized using Unity Jobs and Burst-compiled C#, cutting frame cost from 13 ms to 2 ms (~6× faster).
- Trained **autonomous drones** via physics-based deep **reinforcement learning** (Unity ML-Agents) with depth-camera vision input, performing over 50 million training steps to reach 95% navigation accuracy in dense forest environments.
- Generated missile telemetry datasets for AI targeting and machine-learning research.

Full-Body Avatar Estimation for Samsung XR Devices - CERTH / Samsung Electronics
Unity, Sentis, ONNX Runtime, TensorFlow Lite

- Prototyped AI full-body avatar estimation trained on AMASS motion-capture data, predicting full-body animation in real time from only head and hand tracking.
- Benchmarked inference engines across mobile XR devices, reaching sub-1 ms latency per prediction under mobile CPU/GPU constraints.

Multiplayer VR Automotive Tour over 5G - CERTH
Unity, Netcode for GameObjects, 360-degree video

- Developed a **multiplayer VR** application in Unity that receives a live 360° video stream from a real vehicle over 5G and projects it to VR users in real time.
- Implemented **foveated rendering** and AI-based face anonymization, maintaining visual fidelity while reducing bandwidth usage by up to 40%.

Palimpsisto – 3D Geospatial Platform - CERTH
Unity, Cesium, REST APIs, AI

- Developed a geospatial platform for the archaeological site of Kythnos, funded by the Greek government, streaming photogrammetry as Cesium 3D Tiles through a REST/JSON backend.
- Implemented an in-platform AI assistant and live museum metadata via REST API, so curators can update exhibits dynamically.